

Common BIM/CAD Model Export & Parsing Solutions

A clear, structured overview of the four main industry approaches.

1. glTF / GLB (Lightweight Mesh Standard)

Suitable For

- Web 3D visualization
- Mobile visualization
- Lightweight model delivery
- Game engines (Unity / Unreal)
- Revit/CAD mesh export
- Large model streaming

Not Suitable For

- BIM semantic data
- Detailed property sets
- Engineering quantity takeoff
- Complex relationships

Tech Stack

- Core Format: glTF (JSON) / GLB (binary)
- Geometry: Binary buffers (.bin)
- Compression: Draco (optional)
- Materials: PBR materials
- Scene Structure: Node graph

Typical Use Cases

- Revit → glTF
- CAD → glTF
- WebGL (Three.js / Babylon.js)
- Unity / Unreal import

Summary

- Pros: lightweight, fast, cross-platform
 - Cons: lacks BIM semantics, not suitable for engineering analysis
-

2. IFC (Building Information Exchange Standard)

Suitable For

- BIM data exchange
- Data migration between platforms
- Semantic model reconstruction

- Property inspection
- Material takeoff (BOQ/BOM)
- Spatial structure analysis
- Long-term archiving

Not Suitable For

- High-precision mesh
- Rendering-level geometry
- Lightweight visualization
- Game engines

Tech Stack

- Core Format: IFC STEP / IFC XML
- Geometry: BREP, Extrusion, SweptSolid
- Attributes: Property Sets (Pset)
- Relationships: Rel series (RelContained, RelAggregates, etc.)

Typical Use Cases

- Revit → IFC → Other BIM software
- Government submission
- BIM platform data migration
- Semantic model reconstruction

Summary

- Pros: rich semantics, strong interoperability
 - Cons: geometry simplified, no mesh fidelity
-

3. Custom Binary Mesh + SQLite (Enterprise-Grade Solution)

Suitable For

- Very large models (hundreds of MB to several GB)
- Desktop BIM platforms
- Backend processing pipelines
- High-performance loading
- Streaming and partial loading
- Custom 3D engines

Not Suitable For

- Cross-platform exchange
- Standardized workflows

Tech Stack

- Geometry: Custom binary mesh (.bin / .mesh)
- Indexing: SQLite (R-tree optional)

- Metadata: JSON or SQLite tables
- Compression (optional): zstd, LZ4, Draco
- Serialization (optional): Protobuf, FlatBuffers

Typical Use Cases

- Enterprise BIM platforms
- CAD/BIM backend processing
- High-performance desktop applications
- Custom visualization engines

Summary

- Pros: extremely fast, scalable, flexible
- Cons: requires custom implementation

Final Overview

1. glTF / GLB

- Lightweight mesh format
- Best for Web/Mobile/Game visualization

2. IFC

- Open BIM standard
- Best for data exchange, migration, analysis
- Not suitable for mesh fidelity

3. Custom Binary + SQLite

- Enterprise solution
- Best for huge models, high performance, custom engines
- Flexible and scalable